

Assignment A1

MATH 147

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September 9, 2026

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1 Problem 1

(I) Prove that $\sqrt{3}$ is irrational.

We need to prove a lemma first before we prove the proposition.

Lemma 1. For all $n \in \mathbb{Z}$, if $3 \mid n^2$, then $3 \mid n$.

Proof. Suppose $3 \nmid n$. There are two cases to discuss.

case 1: $n = 3k + 1$

According to the definition of divisibility, we have

$$n = 3k + 1 \quad \exists k \in \mathbb{Z}.$$

Taking the square of both sides, it follows that

$$n^2 = (3k + 1)^2 = 9k^2 + 6k + 1 = 3c + 1 \quad c = 3k^2 + 2k \in \mathbb{Z}.$$

thus, $3 \nmid n^2$.

case 2: $n = 3k + 2$

According to the definition of divisibility, we have

$$\exists k \in \mathbb{Z} \text{ s.t. } n = 3k + 2$$

Taking the square of both sides, it follows that

$$n^2 = (3k + 2)^2 = 9k^2 + 12k + 4 = 3c + 1 \quad c = 3k^2 + 4k + 1 \in \mathbb{Z}.$$

thus, $3 \nmid n^2$. ■

Here is the proof of the question:

Proof. Suppose $\sqrt{3} \in \mathbb{Q}$.

Then, according to the definition of rational numbers,

$$\exists m, n \in \mathbb{Z}, n \neq 0 \quad \text{s.t.} \quad \sqrt{3} = \frac{m}{n}.$$

We assume that m and n have no common factor other than 1. Taking the square of the both sides, we get

$$3 = \frac{m^2}{n^2} \implies m^2 = 3n^2,$$

which implies that $3 \mid m$ according to the lemma 1. According to the definition of divisibility,

$$\exists k \in \mathbb{Z} \quad \text{s.t.} \quad m = 3k.$$

Subbing $m = 3k$ back into $m^2 = 3n^2$, we have

$$\begin{aligned} n^2 &= (3k)^2 / 3 \\ &= 9k^2 / 3 \\ &= 3d, d = k^2 \in \mathbb{Z}. \end{aligned}$$

It means that $3 \mid n$.

Since $3 \mid n$ and $3 \mid m$, it contradicts the assumption that n and m have no common factor other than 1.

Therefore, $\sqrt{3} \in \mathbb{R} \setminus \mathbb{Q}$. ■

(II)